



Linear Algebra & Fourier Analysis

Assignment 02

Total - 100 Marks, Due date: **Monday, September 08**

(You need to answer one question from section-1 and three questions from section-2,3,4. Total 10 questions need to be answered)

1 Eigenvalues and Eigenvectors

1. Consider

$$A = \begin{pmatrix} 2 & -3 & 6 \\ 0 & 5 & -6 \\ 0 & 1 & 0 \end{pmatrix}$$

Find characteristic equations, Eigenvalues, and Eigenvectors. Diagonalize A . Hence, compute A^{100} . (10 Marks)

2. Let C be a real 3×3 matrix such that

- The trace of C is 7,
- The determinant of C is 6,
- One eigenvalue is 1,
- C is diagonalizable over $\mathbb{R}$.

(a) Find the other two eigenvalues.

(b) If C is similar to a diagonal matrix D , write down D explicitly.

(c) Find an example of such a matrix C that is **not symmetric**.

2 Inner Product

1. Let $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T A \mathbf{v}$ where

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad \mathbf{u}, \mathbf{v} \in \mathbb{R}^2.$$

Check whether this defines a valid inner product on $\mathbb{R}^2$. (10 Marks)

2. Let $f(x) = x$, $g(x) = x^2$ defined on $[0, 1]$. Compute their inner product in the space $C[0, 1]$ with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x) dx.$$

Show that $\sin(x)$ and $\cos(x)$ are orthogonal in this inner product space. (10 Marks)

3. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$. Define inner product on $M_{2 \times 2}(\mathbb{R})$ by

$$\langle A, B \rangle = \text{trace}(A^T B).$$

Compute $\langle A, B \rangle$. (10 Marks)

4. Apply the Gram-Schmidt process to the functions $f_1(x) = 1$, $f_2(x) = x$, $f_3(x) = x^2$ over $[0, 1]$ with inner product $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$, and compute the orthonormal set up to f_2 . (10 Marks)
5. Apply Gram-Schmidt to orthogonalize the set $\{(1, 1, 1, 1), (1, 1, -1, -1), (1, -1, -1, 1)\}$ in $\mathbb{R}^4$. Normalize the result to get an orthonormal basis. (10 Marks)
6. Find the orthogonal complement $W^\perp$ using the standard inner product of the subspace,

$$W = \text{Span} \left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} \right\}$$

Hint: $\forall u \in W, \forall v \in W^\perp$ we have $\langle u, v \rangle = 0$. Okay, then consider $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in W^\perp$, now

calculate the inner product and find the subspace. (10 Marks)

3 Fourier series

1. Consider the function,

$$f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

- (a) Determine whether the function is even, odd or neither even nor odd.
 (b) Find the Fourier series expansion of the function $f(x)$.

(10 Marks)

2. Expand $f(x) = \cos(x)$, $0 < x < \pi$ in a Fourier sine series.

(10 Marks)

3. (a) Expand $f(x) = x$, $0 < x < 2$ in a half range series of cosines.

- (b) Using the parseval's identity for this case and hence deduce the value of $S = \sum_{n=1}^{\infty} \frac{1}{n^4}$.

(10 Marks)

4. Expand $f(x) = x^2$, $0 < x < 2\pi$, in a Fourier series if the period is 2π .

(10 Marks)

5. Expand

$$f(x) = \begin{cases} x, & 0 < x < 4 \\ 8 - x, & 4 < x < 8 \end{cases}$$

in a Fourier series of cosines.

(10 Marks)

6. Use euler identity to prove that the complex form of Fourier series can be expressed as

$$\sum_{n=-\infty}^{\infty} c_n e^{\frac{n\pi x i}{L}}$$

(10 Marks)

4 Fourier Transform

1. Find the Fourier cosine transform of e^{-mx} , $x \geq 0$. Hence show that,

$$\int_0^{\infty} \frac{\beta \cos(pv)}{v^2 + \beta^2} dv = \frac{\pi}{2} e^{-p\beta}, p > 0, \beta > 0$$

(10 Marks)

2. Find the Fourier transform of the function,

$$f(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$$

Hence, determine

$$\int_0^\infty \frac{\sin^4 x}{x^2} dx = \frac{\pi}{4}$$

(10 Marks)

3. Find the Fourier transform of the function,

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

Hence, show that

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$$

(10 Marks)

4. Find the Fourier transform of the function,

$$f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$$

Hence, show that

$$\int_0^\infty \frac{\sin(ax) \cos(ax)}{x} dx = \frac{\pi}{4}$$

(10 Marks)

5. Find the Fourier transform of the function,

$$f(x) = \begin{cases} 1 - x^2, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

Hence, evaluate the definite integral

$$\int_0^\infty \frac{x \cos(x) - \sin(x)}{x^3} \cos\left(\frac{x}{2}\right) dx$$

(10 Marks)

6. Find the Fourier transform of the function,

$$f(x) = \begin{cases} 1 - |x|, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

Hence, evaluate the definite integral

$$\int_0^\infty \frac{\sin^2 x}{x^2} dx$$

(10 Marks)

Best of Luck!

MAT216

Assignment - 2

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Section: 22

Eigenvalue and Eigenvector

$$A = \begin{bmatrix} 2 & -3 & 6 \\ 0 & 5 & -6 \\ 0 & 1 & 0 \end{bmatrix}$$

$$(A - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -3 & 6 \\ 0 & 5-\lambda & -6 \\ 0 & 1 & 0-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda) \{ (5-\lambda)(-\lambda) - (-6) \} - (-3) \{ 0 - 0 \} + 6 \{ 0 \} = 0$$

$$\Rightarrow -10\lambda + 2\lambda^2 + 12 + 5\lambda^2 - \lambda^3 - 6\lambda = 0$$

$$\Rightarrow -\lambda^3 + 7\lambda^2 - 16\lambda + 12 = 0$$

$$\lambda = 3, 2, 2$$

now,

$$\lambda = 3,$$

$$\begin{bmatrix} 2-\lambda & -3 & 6 \\ 0 & 5-\lambda & -6 \\ 0 & 1 & 0-\lambda \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & -3 & 6 \\ 0 & 2 & -6 \\ 0 & 1 & -3 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} -1 & -3 & 6 & 0 \\ 0 & 2 & -6 & 0 \\ 0 & 1 & -3 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} -1 & -3 & 6 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 1 & -3 & 0 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|c} -1 & -1 & 0 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$\therefore$ let, $x_3 = t$.

$\therefore x_1 - x_2 = 0$

$x_2 - 3x_3 = 0$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -3t \\ 3t \\ t \end{pmatrix} = t \begin{pmatrix} -3 \\ 3 \\ 1 \end{pmatrix}$$

for $\lambda = 3$, $\begin{pmatrix} -3 \\ 3 \\ 1 \end{pmatrix}$ is an eigen vector.

$\lambda = 2$,

$$\begin{pmatrix} 2-\lambda & -3 & 6 \\ 0 & 5-\lambda & -6 \\ 0 & 1 & -\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -3 & 6 \\ 0 & 3 & -6 \\ 0 & 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 0 & -3 & 6 & 0 \\ 0 & 3 & -6 & 0 \\ 0 & 1 & -2 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 0 & -3 & 6 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\sim \left(\begin{array}{ccc|ccc} 0 & 1 & -2 & 0 & 0 & 0 \\ 0 & -3 & 6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 0 & 1 & -2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_2' = R_2 + 3R_1}$$

$$\lambda_2 - 2\lambda_3 = 0$$

$$\lambda_2 = 2\lambda_3$$

$$\lambda_1 = t_1$$

$$\lambda_3 = t_2$$

$$\begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = \begin{pmatrix} t_1 \\ 2t_2 \\ t_2 \end{pmatrix} = t_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$

for $\lambda = 2$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ are eigenvector.

A is diagonalizable

$$A = \begin{pmatrix} 2 & -3 & 6 \\ 0 & 5 & -6 \\ 0 & 1 & 0 \end{pmatrix}, D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, P = \begin{pmatrix} -3 & 1 & 0 \\ 3 & 0 & 2 \\ 1 & 0 & 1 \end{pmatrix}$$

$$P^{-1} = \begin{pmatrix} 0 & 1 & -2 \\ 1 & 3 & -6 \\ 0 & -1 & 3 \end{pmatrix}$$

$$\text{for } A = PDP^{-1}$$

$$\therefore A^{100}$$

$$A^{100} = PD^{100}P^{-1}$$

$$\Rightarrow \begin{pmatrix} -3 & 1 & 0 \\ 3 & 0 & 2 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3^{100} & 0 & 0 \\ 0 & 2^{100} & 0 \\ 0 & 0 & 2^{100} \end{pmatrix} \begin{pmatrix} 0 & 1 & -2 \\ 1 & 3 & -6 \\ 0 & -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 1 & 0 \\ 3 & 0 & 2 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 6 & 3^{100} & -2 \cdot 3^{100} \\ 2^{100} & 3 \cdot 2^{100} & -6 \cdot 2^{100} \\ 0 & -2^{100} & 3 \cdot 2^{100} \end{pmatrix}$$

$$= \begin{pmatrix} 2^{100} & -3 \cdot 3^{100} + 3 \cdot 2^{100} & 6 \cdot 3^{100} - 6 \cdot 2^{100} \\ 0 & 3 \cdot 3^{100} - 2 \cdot 2^{100} & -6 \cdot 3^{100} + 6 \cdot 2^{100} \\ 0 & 3^{100} - 2^{100} & -2 \cdot 3^{100} + 3 \cdot 2^{100} \end{pmatrix} \text{ Ans}$$

Inner product

3] $A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 \end{bmatrix}, B = \begin{bmatrix} 6 & 1 \\ 1 & 2 \end{bmatrix}$

$$\langle A, B \rangle = \text{trace}(A^T B)$$

$$A^T B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}^T \begin{bmatrix} 6 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 6 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 3 & 8 \end{bmatrix}$$

$$\text{Trace}(A^T B) = \text{Trace} \begin{bmatrix} 6 & 1 \\ 3 & 8 \end{bmatrix}$$

$$= 8 \text{ Ans}$$

$$6) W = \text{Span} \left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} \right\}$$

Let,

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in W^\perp \quad \left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \right\} = 0 \Rightarrow x_1 + 3x_2 + 0 = 0 \quad \dots (i)$$

$$\left\{ \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \right\} = 0 \Rightarrow 2x_1 + x_2 + 4x_3 = 0 \quad \dots (ii)$$

$$\left(\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 2 & 1 & 4 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & -5 & 4 & 0 \end{array} \right) \quad (R_2' = R_2 - 2R_1)$$

$$\Rightarrow x_1 + 3x_2 = 0 \Rightarrow x_1 = -3x_2$$

$$\Rightarrow -5x_2 + 4x_3 = 0 \Rightarrow x_2 = \frac{4}{5}x_3$$

$$\text{Let, } x_3 = t.$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -\frac{12}{5}t \\ \frac{4}{5}t \\ t \end{pmatrix} = \frac{t}{5} \begin{pmatrix} -12 \\ 4 \\ 5 \end{pmatrix}$$

$$W^\perp = \text{Span} \left\{ \begin{pmatrix} -12 \\ 4 \\ 5 \end{pmatrix} \right\}$$

1] let, $\langle u, v \rangle = u^T A \cdot v$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, u, v \in \mathbb{R}^2$$

Solve

$$u^T A v = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}^T \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$= \begin{pmatrix} u_1 & u_2 \end{pmatrix}_{1 \times 2} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}_{2 \times 2} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}_{2 \times 1}$$

$$= \begin{pmatrix} 2u_1 & 3u_2 \end{pmatrix}_{1 \times 2} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}_{2 \times 1} = (2u_1 v_1 + 3u_2 v_2)_{1 \times 1}$$

$$\langle u, v \rangle = (2u_1 v_1 + 3u_2 v_2)$$

A₁ : $\langle \vec{u}, \vec{u} \rangle \geq 0$ for all $\vec{u} \in V$

$$\langle \vec{u}, \vec{u} \rangle = 2u_1^2 + 3u_2^2 \geq 0 \quad (\text{satisfied})$$

A₂ : $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = 0$.

for $\vec{u} = 0$ $\langle \vec{u}, \vec{u} \rangle = 2u_1^2 + 3u_2^2 = 0$, satisfied.

A₃ : $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ for all
 $\vec{u}, \vec{v}, \vec{w} \in V$
 $\vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$

$$\begin{aligned}
 \langle \vec{u} + \vec{v}, \vec{w} \rangle &= 2(u_1 + v_1)w_1 + 3(u_2 + v_2)w_2 \\
 &= 2u_1w_1 + 2v_1w_1 + 3u_2w_2 + 3v_2w_2 \\
 &= (2u_1w_1 + 2u_2w_2) + (2v_1w_1 + 3v_2w_2) \\
 &= \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle \quad (\text{satisfied})
 \end{aligned}$$

A4: $\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$

$$\begin{aligned}
 \langle \lambda \vec{u}, \vec{v} \rangle &= 2\lambda u_1 v_1 + 3\lambda u_2 v_2 \\
 &= \lambda (2u_1 v_1 + 3u_2 v_2) \\
 &= \lambda \langle \vec{u}, \vec{v} \rangle \quad (\text{satisfied})
 \end{aligned}$$

A5: $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ for all $\vec{u}, \vec{v} \in V$

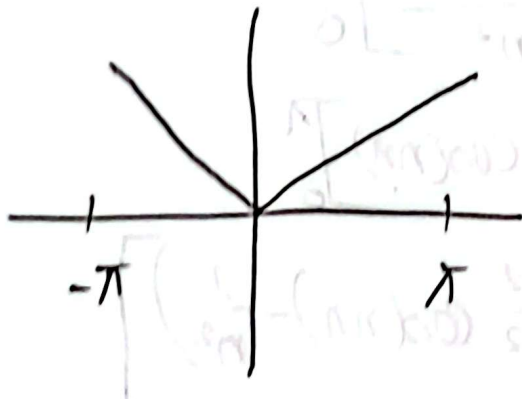
$$\begin{aligned}
 \langle \vec{u}, \vec{v} \rangle &= 2u_1 v_1 + 3u_2 v_2 \\
 &= 2v_1 u_1 + 3v_2 u_2 \\
 &= \langle \vec{v}, \vec{u} \rangle \quad (\text{satisfied})
 \end{aligned}$$

$\therefore \langle \vec{u}, \vec{v} \rangle$ is a valid inner product

Fourier series

$$1) f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

a)



even function.

for $-\pi < x < 0$

$$f(x) = -x$$

$$f(-\pi) = \pi$$

$$f(-1) = 1$$

for $0 < x < \pi$

$$f(x) = x$$

$$f(1) = 1$$

$$f(\pi) = \pi$$

$$\begin{aligned} \underline{b)} \quad f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) \quad \left| \begin{array}{l} 2L = 2\pi \\ L = \pi \end{array} \right. \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) \end{aligned}$$

$$\therefore a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{\pi^2}{2} - 0 \right] = \pi$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos(nx) dx.$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos(nx) dx.$$

x	$\cos(nx)$
1	$\frac{\sin(nx)}{n}$
0	$-\frac{\cos(nx)}{n^2}$

$$= \frac{2}{\pi} \left[x \frac{\sin(nx)}{n} - (1) \frac{-\cos(nx)}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\left(\frac{\pi}{n} \sin(n\pi) - 0 \right) + \frac{1}{n^2} (\cos(n\pi) - \frac{1}{n^2}) \right]$$

$$= \frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] = \frac{2((-1)^n - 1)}{\pi n^2}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \frac{2((-1)^n - 1)}{\pi n^2} \cos(nx)$$

The fourier series is, $f(x) =$

$$= \frac{\pi}{2} + \frac{2(-1) - 1}{\pi} \cos x + \frac{2((-1)^2 - 1)}{4\pi} \cos 2x + \dots$$

Ans

Q2) $f(x) = \cos x$, $0 < x < \pi$ in sine series.

Ans

$$L = \pi$$

$$f(x) = \sum_{n=1}^{\infty} (b_n \sin(nx))$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx) dx \quad (n \neq 1)$$

$$= \frac{2}{\pi} \int_0^{\pi} (\cos x) \sin(nx) dx.$$

$$= \frac{1}{\pi} \int_0^{\pi} 2(\cos x) \sin(nx) dx.$$

$$= \frac{1}{\pi} \int_0^{\pi} 2 \sin(nx) \cos(x) dx.$$

$$= \frac{1}{\pi} \int_0^\pi [\sin(nx+x) + \sin(nx-x)] dx.$$

$$= \frac{1}{\pi} \left[\frac{-\cos(nx+x)}{n+1} + \frac{-\cos(nx-x)}{n-1} \right]_0^\pi$$

$$= \frac{1}{\pi} \left[\frac{\cos(nx+x)}{n+1} + \frac{\cos(nx-x)}{n-1} \right]_0^\pi$$

$$= \frac{1}{\pi} \left[\left(\frac{1}{n+1} + \frac{1}{n-1} \right) - \left(\frac{\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1} \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{1}{n+1} + \frac{1}{n-1} - \frac{(-1)^{n+1}}{n+1} - \frac{(-1)^{n-1}}{n-1} \right]$$

$$= \frac{1}{\pi} \left[\frac{2n}{n^2-1} - \frac{(-1)^n \cdot (-1)^1}{n+1} - \frac{(-1)^n \cdot (-1)^{-1}}{n-1} \right]$$

$$= \frac{1}{\pi} \left[\frac{2n}{n^2-1} + \frac{(-1)^n}{n+1} + \frac{(-1)^n}{n-1} \right]$$

$$= \frac{1}{\pi} \left[\frac{n}{n^2-1} + \frac{(-1)^n}{n+1} + \frac{(-1)^n}{n-1} \right]$$

$$\Rightarrow \frac{1}{\pi} \left[\frac{n}{n^2-1} + \frac{n \cdot (-1)^n}{n^2-1} \right]$$

$$= \frac{n(1+(-1)^n)}{\pi(n^2-1)}$$

$$b_1 = \frac{2}{\pi} \int_0^{\pi} \cos x \sin x \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin 2x \, dx$$

$$= \frac{1}{\pi} \left[\frac{-\cos 2x}{2} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\cos 2x}{2} \right]_{\pi}^0$$

$$= \frac{1}{\pi} \left[\frac{1}{2} - \frac{1}{2} \right]$$

$$= 0$$

Q11) $f(x) = x^2$, $0 \leq x \leq 2\pi$, where period is 2π , Fourier expansion,
Ans

$$2L = 2\pi$$

$$L = \pi$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + \sin\left(\frac{n\pi x}{L}\right) \right)$$
$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + \sin(nx))$$

$$a_n = \frac{1}{L} \int_0^{2\pi} f(x) \cos(nx) dx, \quad b_n = \frac{1}{L} \int_0^{2\pi} f(x) \sin(nx) dx$$

$$\therefore a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 dx = \frac{1}{\pi} \left[\frac{x^3}{3} \right]_0^{2\pi} = \frac{1}{\pi} \left(\frac{8\pi^3}{3} - 0 \right)$$
$$= \frac{8\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cdot \cos(nx) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \cos(nx) dx$$

$$= \frac{1}{\pi} \left[(x^2) \left(\frac{\sin nx}{n} \right) - (2x) \left(\frac{-\cos nx}{n^2} \right) + 2 \left(-\sin \frac{nx}{n^3} \right) \right]_0^{2\pi}$$

u	v
x^2	$\cos nx$
$2x$	$\frac{\sin nx}{n}$
$+ 2$	$-\frac{\cos nx}{n^2}$
0	$-\frac{\sin nx}{n^3}$

$$= \frac{1}{\pi} \left[x^2 \frac{\sin nx}{n} + 2x \frac{\cos nx}{n^2} - 2 \frac{\sin nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[4\pi^2 \frac{\sin(2n\pi)}{n} + 4\pi \frac{\cos(2n\pi)}{n^2} - 2 \frac{\sin(2n\pi)}{n^3} \right] - (0 + 0 - 0)$$

$$= \frac{1}{\pi} \left(4\pi \frac{(-1)^{2n}}{n^2} \right) \rightarrow \text{even}$$

$$= 4 \frac{1}{n^2} = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \sin(nx) dx$$

$$= \frac{1}{\pi} \left[(x^2) \left(\frac{-\cos nx}{n} \right) - (2x) \left(\frac{-\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right]_0^{2\pi}$$

u	v
x^2	$\sin nx$
$2x$	$-\frac{\cos nx}{n}$
$+ 2$	$-\frac{\sin nx}{n^2}$
$- 0$	$\frac{\cos nx}{n^3}$

$$= \frac{1}{\pi} \left[-x^2 \frac{\cos nx}{n} + 2x \frac{\sin nx}{n^2} + 2 \frac{\cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\left(-4\pi^2 \frac{\cos(2n\pi)}{n} + 4\pi \frac{\sin(2n\pi)}{n^2} + 2 \frac{\cos(2n\pi)}{n^3} \right) - \left(0 + 0 + \frac{2}{n^3} \right) \right]$$

$$= \frac{1}{\pi} \left[-4\pi^2 \cdot \frac{(-1)^{2n}}{n} + 2 \frac{(-1)^{2n}}{n^3} - \frac{2}{n^3} \right]$$

$$= \frac{1}{\pi} \left[-\frac{4\pi^2}{n} + \frac{2}{n^3} - \frac{2}{n^3} \right] = -\frac{4\pi}{n}$$

$$\therefore a_0 = \frac{8\pi^2}{3}, \quad a_n = \frac{4}{n^2}, \quad b_n = -\frac{4\pi}{n}$$

Fourier Transform

$$4. f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases} \Rightarrow (-a < x < a)$$

evaluate,

$$\int_0^{\infty} \frac{\sin(ax) \cos(ax)}{x} dx$$

Ans

F.T

$$\begin{aligned} F(w) &= \int_{-\infty}^{\infty} f(x) e^{-iwx} dx \\ &= \int_{-a}^a 1 \cdot e^{-iwx} dx \\ &= \left[\frac{e^{-iwx}}{-i} \right]_{-a}^a \\ &= \frac{e^{-iaw} - e^{iaw}}{-i} \\ &= \frac{e^{iaw} - e^{-iaw}}{i} \\ &= \frac{2i \sin(aw)}{i} \\ &= \frac{2 \sin(aw)}{w} \end{aligned}$$

I.F.T

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(w) e^{iwx} dw$$

$$\Rightarrow f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(aw)}{w} e^{iwx} dw$$

$$\Rightarrow f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(aw)}{w} (\cos wx + i \sin wx) dw$$

$$\Rightarrow f(x) + i0 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(aw)}{w} \cos(wx) dw + i \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(aw)}{w} \sin(wx) dw$$

$$\text{now, } \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(aw)}{w} \cos(wx) dw = f(x)$$

$$\Rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\sin(aw)}{w} \cdot \cos(aw) dw = f(a) \quad [x=a]$$

$$\Rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\sin(aw)\cos(aw)}{w} dw = \frac{1+0}{2}$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\sin(aw)\cos(aw)}{w} dw = \frac{\pi}{2}$$

$$\Rightarrow 2 \int_0^{\infty} \frac{\sin(aw)}{w} \cdot \cos(aw) dw = \frac{\pi}{2}$$

$$\Rightarrow \int_0^{\infty} \frac{\sin(aw)}{w} \cdot \cos(aw) dw = \frac{\pi}{4}$$

Ans

Q5] $f(x) = \begin{cases} 1-x^2, & |x| < 1 \\ 0, & |x| > 1 \end{cases} \quad (-1 < x < 1)$

evaluate,

$$\int_0^{\infty} \left(\frac{x \cos x \sin x}{x^3} \right) \cos\left(\frac{x}{2}\right) dx$$

Ans

F.T

$$F(w) = \int_{-\infty}^{\infty} f(x) e^{-iwx} dx$$

$$= \int_{-1}^1 (1-x^2) e^{-iwx} dx$$

$$= \left[(1-x^2) \frac{e^{-iwx}}{-iw} - (-2x) \frac{e^{-iwx}}{i^2 w^2} + (-2) \frac{e^{-iwx}}{i^3 w^3} \right]_{-1}^1$$

$$= \left[(1-x^2) \frac{e^{-iwx}}{-i\omega} - 2x \frac{e^{-iwx}}{\omega^2} - 2 \frac{e^{-iwx}}{i\omega^3} \right]_{-1}^1$$

$1-x^2$	e^{-iwx}
$2x$	$\frac{e^{-iwx}}{-i\omega}$
-2	$\frac{e^{-iwx}}{i^2\omega^2}$
0	$\frac{e^{-iwx}}{-i^3\omega^3}$

$$= \left[(1-x^2) \frac{e^{-iwx}}{+i\omega} + 2x \frac{e^{-iwx}}{\omega^2} + 2 \frac{e^{-iwx}}{i\omega^3} \right]_{-1}^1$$

$$= \left(0 \frac{e^{i\omega}}{i\omega} - 2 \frac{e^{i\omega}}{\omega^2} + 2 \frac{e^{i\omega}}{i\omega^3} \right) - \left(0 + 2 \frac{e^{-i\omega}}{\omega^2} + 2 \frac{e^{-i\omega}}{i\omega^3} \right)$$

$$= -2 \frac{e^{i\omega}}{\omega^2} - 2 \frac{e^{-i\omega}}{\omega^2} + 2 \frac{e^{i\omega}}{i\omega^3} - 2 \frac{e^{-i\omega}}{i\omega^3}$$

$$= -2 \frac{e^{i\omega} + e^{-i\omega}}{\omega^2} + 2 \frac{e^{i\omega} - e^{-i\omega}}{i\omega^3}$$

$$= -2 \frac{2\cos\omega}{\omega^2} + 2 \frac{2i\sin\omega}{i\omega^3}$$

$$= \frac{-4\cos\omega}{\omega^2} + \frac{4\sin\omega}{\omega^3}$$

$$= \frac{4\sin\omega - 4\omega\cos\omega}{\omega^3}$$

I. F. T

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

$$\therefore f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4\sin\omega - 4\omega\cos\omega}{\omega^3} e^{i\omega x} d\omega$$

$$\Rightarrow f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin w - w \cos w}{w^3} (\cos wx + i \sin wx) dw$$

$$\therefore f(x) + i0 = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w \cos w}{w^3} (\cos wx) dw + i \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w \cos w}{w^3} \sin wx dw$$

$$\therefore \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w \cos w}{w^3} (\cos wx) dw = f(x)$$

$$\Rightarrow \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w \cos w}{w^3} \left(\cos \frac{w}{2} \right) dw = f\left(\frac{1}{2}\right) \quad \left[x = \frac{1}{2} \right]$$

$$\Rightarrow \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w \cos w}{w^3} \left(\cos \frac{w}{2} \right) dw = \frac{\pi}{2} \left(1 - \left(\frac{1}{2} \right)^2 \right)$$

$$\Rightarrow 2 \int_0^{\infty} \frac{\sin x - x \cos x}{x^3} \left(\cos \frac{x}{2} \right) dx = \frac{\pi}{2} \cdot \frac{3}{4}$$

$$= 2 \int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \left(\frac{x}{2} \right) dx = \frac{3\pi}{8}$$

$$\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx = -\frac{3\pi}{12}$$

Any

$$f(x) = \begin{cases} 1-|x|, & |x| < 1 \\ 0, & |x| > 1 \end{cases} \quad (-1 < x < 1)$$

evaluate,

$$\int_0^{\infty} \frac{\sin^2 x}{x} dx$$

→

F.T

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \int_{-1}^1 (1-|x|) e^{-i\omega x} dx$$

$$= \int_{-1}^0 (1-(-x)) e^{-i\omega x} dx + \int_0^1 (1-(x)) e^{-i\omega x} dx$$

$$= \left[(1+x) \frac{e^{-i\omega x}}{-i\omega} - \frac{e^{-i\omega x}}{i^2 \omega^2} \right]_{-1}^0 + \left[(1-x) \frac{e^{-i\omega x}}{-i\omega} - (-1) \frac{e^{-i\omega x}}{i^2 \omega^2} \right]_0^1$$

$$= \left[(1+x) \frac{e^{-i\omega x}}{-i\omega} - \frac{e^{-i\omega x}}{i^2 \omega^2} \right]_{-1}^0 + \left[(1-x) \frac{e^{-i\omega x}}{-i\omega} - \frac{e^{-i\omega x}}{\omega^2} \right]_0^1$$

$$= \left[(1+x) \frac{e^{-i\omega x}}{-i\omega} - \frac{e^{-i\omega x}}{i^2 \omega^2} \right]_{-1}^0 + \left[(1-x) \frac{e^{-i\omega x}}{i\omega} + \frac{e^{-i\omega x}}{\omega^2} \right]_0^1$$

$$= \left(1 \cdot \frac{1}{-i\omega} + \frac{1}{\omega^2} \right) - \left(0 + \frac{e^{i\omega}}{\omega^2} \right) + \left(1 \cdot \frac{1}{i\omega} + \frac{1}{\omega^2} \right) - \left(0 + \frac{e^{-i\omega}}{\omega^2} \right)$$

$$= \frac{2}{\omega^2} - \frac{e^{i\omega} + e^{-i\omega}}{\omega^2}$$

$$= \frac{2}{\omega^2} - \frac{2 \cos \omega}{\omega^2}$$

$$= \frac{2(1 - \cos \omega)}{\omega^2}$$

$$= \frac{2(1 - \cos \omega)}{\omega^2}$$

$$= \frac{2}{\omega^2} 2 \sin^2 \left(\frac{\omega}{2} \right)$$

$$= \frac{4}{\omega^2} \sin^2 \left(\frac{\omega}{2} \right)$$

I. F. T

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(\frac{\omega}{2})}{\omega^2} e^{i\omega x} d\omega$$

$$\Rightarrow f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(\frac{\omega}{2})}{\omega^2} d\omega$$

$$\Rightarrow 1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(\frac{\omega}{2})}{\omega^2} d\omega$$

$$\Rightarrow 1 = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2(x)}{(2x)^2} 2dx$$

$$\Rightarrow 1 = \frac{2}{\pi} \cdot \frac{2}{4} \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx = \pi$$

$$\Rightarrow 2 \int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \pi$$

$$\therefore \int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$$